

Aberrant salience mediates the interplay between emotional abuse and positive symptoms in schizophrenia

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ABSTRACT

Introduction: Childhood trauma and adversities (CTA) and aberrant salience (AS) have a pivotal role in schizophrenia development, but their interplay with psychotic symptoms remains vague. We explored the mediation performed by AS between CTA and psychotic symptomatology in schizophrenia.

Methods: We approached 241 adults suffering from schizophrenia spectrum disorders (SSDs), who have been in the unit for at least 12 consecutive months, excluding the diagnosis of dementia, and recent substance abuse disorder, and cross-sectional evaluated through the Aberrant Salience Inventory (ASI), Childhood Trauma Questionnaire Short-Form (CTQ-SF), and Positive and Negative Symptom Scale (PANSS). We tested a path-diagram where AS mediated the relationship between CTA and psychosis, after verifying each measure one-dimensionality through confirmatory factor analysis.

Results: The final sample comprised 222 patients (36.9% female), with a mean age of 42.4 ($\pm$ 13.3) years and an average antipsychotic dose of 453.6 ($\pm$ 184.2) mg/day (chlorpromazine equivalents). The mean duration of untreated psychosis was 1.8 ($\pm$ 2.0) years while the mean onset age was 23.9 ($\pm$ 8.2) years. Significant paths were found from emotional abuse to ASI total score (β = 0.39; p < .001) and from ASI total score to PANSS positive (β = 0.17; p = .019). Finally, a statistically significant indirect association was found from emotional abuse to PANSS positive mediated by ASI total score (β = 0.06; p = .041; CI 95% [0.01, 0.13]).

Conclusion: Emotional abuse has an AS-mediated effect on positive psychotic symptomatology. AS evaluation could allow a better characterization of psychosis as well as explain the presence of positive symptoms in adults with SSDs who experienced CTA.

1. Introduction

Salience detection represents an attentional process wherein individuals allocate their perceptual and cognitive resources to identify and ascribe significance to the most pertinent environmental stimuli within a given context [1,2]. The dysfunctional functioning of this process, known as aberrant salience (AS), assumes a pivotal role in the genesis and persistence of psychotic symptoms, specifically delusions, and hallucinations, contributing to distortions in reality perception thus causing impairment [3–5]. In this vein, it has been suggested that

psychosis may stem from an irregular ascription of significance to both external objects and internal representations [6,7]. Kapur further posited that this misattribution of meaning could be attributed to heightened dopaminergic transmission in the brain [6]. Therefore, disruptions in dopaminergic transmission can give rise to anomalous patterns of reward and aversive signaling [8]. Consequently, the blockade of the dopamine D₂ receptor could potentially mitigate the motivational salience and ameliorate positive symptoms [9,10]. However, in the last few years, the purely dopaminergic hypothesis of schizophrenia has been criticized and evaluated as extremely reductionistic. Indeed,

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expanding upon this hypothesis through an evolutionary lens, we acknowledge the significant role of stressful events resulting from various factors such as urban living, migrant status, male gender, low socioeconomic status, subjective social standing, and adverse childhood experiences, thus connecting the dots to the evolutionary origins of psychosis. Therefore, environmental mismatch emerges as a pivotal, but not exclusive, evolutionary mechanism, shedding light on the perplexing persistence of psychosis into an unified account of AS in psychosis [11].

Following this flow, within the causal chain between early risk factors and schizophrenia spectrum disorders (SSDs) [12], a wealth of evidence suggests that childhood trauma and adversities (CTA) significantly impact the development, clinical progression, and long-term outcomes of SSDs as an influential environmental factor [12–14].

The onset, frequency, and severity of psychotic symptoms in schizophrenia, when considered alongside psychosocial and biological protective/risk factors, may probabilistically influence the individual's progression from subtle pre-clinical experiential anomalies to the development of full-fledged psychosis [15,16], as well as determine the different staging of disease, response to therapy and possibilities for rehabilitation and functional recovery [17]. Thus, AS could potentially serve as the starting point on this continuum, characterized by the abnormal assignment of significance to otherwise neutral or inconspicuous stimuli, thereby catalyzing the onset (and consolidation) of cognitive-perceptual positive symptoms [18,19], and CTA may operate as a breaking point of this symptom spectrum during fundamental phases of physical and personal growth [20,21].

This concept escapes current categorical diagnostic models. Indeed, although the new International Classification of Diseases version 11 (ICD-11) developed by the World Health Organization (WHO) redefined schizophrenia, aligning more closely with DSM-5 by removing Schneider's first-rank symptoms and altering specifiers, discrepancies with the assessment of dimensional psychotic symptoms such as AS persist [22,23].

To the best of our knowledge, if the relationship between both CTA and AS with the development, maintenance, and severity of psychotic symptoms in schizophrenia has been well-established, their interconnection and mediating pathways have been only explored in healthy subjects [24,25], but there remains a gap in research delving into the interplay between these two dimensions and their direct or indirect impact on the onset of schizophrenia [26,27]. An enhanced understanding of the interplay between CTA and AS hindering psychotic symptoms is instrumental in characterizing patients at higher risk, tailoring therapeutic management, and improving the disorder prognosis [28].

Hence, we hypothesize that the presence of AS trait and its constituent domains may mediate the relationship between experienced CTA and the severity and clinical manifestations of psychosis during adulthood in individuals with SSDs, notwithstanding the undergoing antipsychotic treatment.

Therefore, the main objective of this research was to investigate the potential relationship between CTA, AS, and psychotic symptoms in individuals suffering from SSDs. Furthermore, our second goal will be to investigate which dimension of CTA may have a greater impact on mediating symptoms of psychosis.

2. Materials and methods

2.1. Participants and procedure

Participants were recruited at the Psychiatry Unit of the University Hospital of Catanzaro (Italy) from July 2020 to July 2023. They were deemed eligible if they fulfilled the following criteria: 1) aged between 18 and 65 years and being able to provide the informed consent form; 2) diagnosed with schizophrenia (SCZ) or schizoaffective disorder by a senior psychiatrist according to the DSM-5 diagnostic criteria [29]; and

3) have been in the unit for at least 12 consecutive months.

Clinical diagnosis was performed using the Structured Clinical Interview for DSM-5 (SCID-5-CV) [30] and according to DSM-5 criteria [29] by an experienced psychiatrist trained in the administration of neuropsychiatric examinations.

Participants were excluded if fulfilling any of the following criteria: 1) diagnosis of dementia, intellectual disability, or other severe medical conditions associated with secondary psychiatric symptoms potentially biasing the assessment; 2) diagnosis of substance abuse disorder in the last six months; 3) inability to complete the assessment due to conditions such as speech impairments or a lack of knowledge of the Italian language; and 4) invalid informed consent to the study procedures.

Before collecting any data, the study protocol was submitted to and approved by the local Ethical Committee of University Hospital Mater Domini of Catanzaro (Italy), 'Regione Calabria, sezione Area Centro' (n. 191/2020). The study procedures and protocol have been carried out according to the ethical principles outlined in the latest version of the Helsinki Declaration [31]. All patients signed a written informed consent according to the ethical committee's guidelines before any data was collected.

A statistical power analysis was not performed for the sample size estimation, but the adherence was to a rule of thumb suggesting that for every parameter employed in the model, approximately 10 participants were required [32].

2.2. Measures

2.2.1. Clinical and sociodemographic characteristics

An ad hoc schedule was administered to assess the demographic and clinical characteristics of the sample. Collected data included psychiatric familiarity, age at onset of illness, duration of untreated psychosis (DUP), previous psychiatric hospitalizations, and antipsychotic therapy. The daily dose of prescribed antipsychotic drugs was converted to the equivalent daily dose of chlorpromazine, based on the international consensus [33].

2.2.2. Assessment

Participants completed the following assessment:

- The *Aberrant Salience Inventory (ASI)* [34], a self-administered test, consisting of 29 items with a dichotomous yes/no response, that assesses individual aberrant salience and psychotic proneness. The scale investigates the presence of five factors: feelings of increased significance (7 items), sense sharpening (5 items), impending understanding (5 items), heightened emotionality (6 items), and heightened cognition (6 items). The global measure of aberrant salience severity is obtained from the sum of all the factors (ranging from 0 to 29) and was used for the analysis ($M = 17.5 \pm 7.0$ SD). The Italian version of the ASI [35] used in this study confirmed the good psychometric properties of the scale. Kuder-Richardson 20 is equivalent to Cronbach's alpha for dichotomous data, and, in the present study, it was 0.90.
- The *Childhood Trauma Questionnaire Short-Form (CTQ-SF)* [36], is a self-administered test consisting of 28 items with a Likert-type response format ranging from 1 (*never*) to 5 (*very often*). The CTQ-SF can be articulated in 5 subscales: emotional abuse ($M = 10.9 \pm 4.7$ SD), physical abuse ($M = 7.9 \pm 3.3$ SD), sexual abuse ($M = 6.6 \pm 3.2$ SD), emotional neglect ($M = 14.8 \pm 5.9$ SD), and physical neglect ($M = 10.1 \pm 3.9$ SD). Their Cronbach alpha and McDonald omega were respectively: physical abuse 0.81 and 0.82; emotional abuse 0.89 and 0.88; sexual abuse 0.89; emotional neglect 0.91 and physical neglect 0.84.
- The *Positive and Negative Symptom Scale (PANSS)* [37], a standardized clinical interview rating the presence and severity of positive and negative symptoms, as well as general psychopathology, for patients with schizophrenia within the past week. The instrument

consists of 30 items with a Likert-type response format ranging from 1 (*absent*) to 7 (*extreme*): the positive ($M = 17.5 \pm 7.5$ SD) and negative subscales consist of seven items each (score: 7–49), whereas the general psychopathology subscale consists of 16 items (score: 16–122). In this study, we considered the positive PANSS subscales according to Wallwork and colleagues' model (2012) [38]. In this model the four items of the PANSS positive were: delusions (P1), hallucinatory behavior (P3), grandiosity (P5), and unusual thought content (G9). The Cronbach's alpha and McDonald's omega were 0.85 for PANSS positive.

All the measures were administered in one session by the same-trained practitioner in this order: socio-demographic data collection, PANSS, ASI, CTQ, and the whole assessment took around 60 min to complete.

2.3. Statistical analysis

In line with the recommendations of Widaman and Revelle (2023) [39], according to whom “*once the dimensional structure of a set of items is verified, sum scores often have a solid psychometric basis and therefore are frequently quite adequate for psychological research*”, we verified the assumption of one-dimensionality through confirmatory factor analysis (CFA) for each measure used. We utilized a covariance matrix of the observed variables in our CFA analysis.

A second-order Confirmatory Factor Analysis (CFA), employing the Diagonally Weighted Least Squares (DWLS) estimator, was conducted to examine the original factor structure of the ASI scale and validate the suitability of a total score. We opted for the DWLS estimator because the items of the ASI are dichotomous. Then, we tested a one-dimensional factor structure for PANSS positive and for CTQ subscales using the Maximum likelihood with robust standard errors (MLR) estimator which is commonly employed for structural equation models when dealing with continuous observed data that depart from a multivariate normal distribution. The relative chi-square (χ^2/df), the Tucker-Lewis Index (TLI), the Comparative Fit Index (CFI), the Root Mean Square Error of Approximation (RMSEA), and the Standardized Root Mean Squared Residual (SRMR) were used to assess the goodness of fit of data for the proposed model. For TLI and CFI, values of 0.90 and above were considered adequate, whereas values of 0.95 or above were considered very good; for RMSEA values of 0.08 and below were considered adequate, and 0.05 or less very good; for SRMR a cut-off value close to 0.08 was considered adequate. Values of $\chi^2/df < 3.0$ are good and those < 2.0 are very good. The levels of these indices were evaluated according to the recommendations of Hu and Bentler [40].

According to the research question, the magnitude of association between variables (CTQ subscales, ASI global score, PANSS positive total score) and the mediational path was examined. We tested if the variables of interest (the dependent, the independent, and the mediator variables) have a linear relationship, which can be checked with a correlation matrix and scatterplots.

Finally, the model tested both the direct and indirect (via aberrant salience) effect of childhood trauma on positive schizophrenia symptomatology.

All the analyses were conducted in R (version 4.3.1, [41]) within R Studio (version 2023.6.1.524, [42]). We used the *lavaan* (version 0.6–16, [43]), the *psych* (version 2.3.6, [44]) and the *semTools* (version 0.5–6, [45]) packages.

3. Results

3.1. Sample characteristics

Out of the initial cohort ($N = 241$) nineteen (7.9%) individuals were deemed ineligible for inclusion in the study due to various reasons as described in Fig. 1.

Consequently, the ultimate dataset consisted of 222 participants, 189 affected by SCZ, and 33 by schizoaffective disorder, who were sequentially recruited during this 3-year practical clinical investigation. Furthermore, the minimal rate of patient attrition (i.e., 7.9%), coupled with an impressive complete response rate exceeding 92.1%, indicates a strong acceptance of the evaluation methodologies by the participants. Therefore, only the patients who accepted to participate in the research protocol, gave their informed consent, and completed the evaluation have been included in the analysis, and no missing data have been reported in the participants' socio-demographic information or the assessment.

Table 1 presents the socio-demographic and clinical characteristics of the sample. The final sample comprised 222 patients of which 140 (63.1%) identified themselves as males and 82 (36.9%) as female; 219 participants were Caucasian (98.6%), and 3 were black (1.4%). The mean age of the participants was 42.4 ± 13.3 years and the mean of years of education was 11.8 ± 3.7 . Most of the participants (68.9%) reported no familial history of SSD and the average dose of antipsychotics was 453.6 ± 184.2 mg/day (chlorpromazine equivalents), with 191 (86.0%) of patients previously exposed to the antipsychotic treatment. The mean DUP was 1.8 ± 2.0 years while the mean age of onset of the disorder was 23.9 ± 8.2 years. Finally, the majority of the sample was single (82.9%) and unemployed (46.4) and the mean of previous psychiatric hospitalizations was 1.4 ± 2.3 .

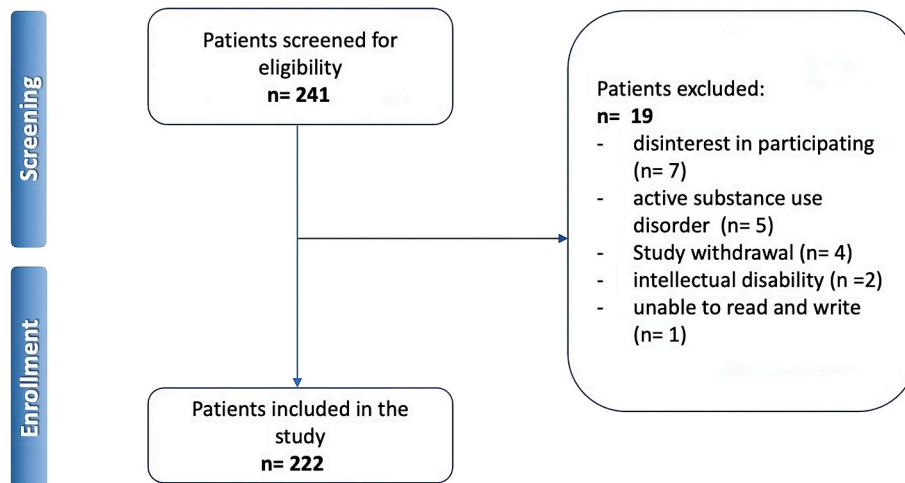


Fig. 1. Sample flow-chart.

Table 1
Socio-demographic and clinical characteristics of the sample.

		N = 222	
		Fr	%
Mean age (years) ^a		42.4	(13.3)
Gender	Male	140	(63.1)
	Female	82	(36.9)
Civil status	Married	25	(11.3)
	Divorced	13	(5.9)
	Single	184	(82.9)
Ethnicity	Caucasian	219	(98.6)
	Black	3	(1.4)
Education (years) ^a		11.8	(3.7)
Employment	Employed	38	(17.1)
	Unemployed	103	(46.4)
	On pension	19	(8.6)
	Unpaid activity	44	(19.8)
	Invalid/Retired	18	(8.1)
Diagnosis	Schizophrenia	189	(85.1)
	Schizoaffective Disorder	33	(14.9)
Familiarity for psychosis	Yes	69	(31.1)
	No	153	(68.9)
Previous antipsychotic treatment	Yes	191	(86.0)
	No	31	(14.0)
Age at onset (years) ^a		23.9	(8.2)
Duration of Untreated Psychosis (years) ^a		1.8	(2.0)
Previous psychiatric hospitalizations ^a		1.4	(2.3)
Daily equivalent dose of chlorpromazine (mg) ^a		453.6	(184.2)

^a Data are expressed as means and (standard deviation).

3.2. Confirmatory factor analysis of each measure

As shown in Table 2, the second-order CFA for the ASI scale yielded acceptable results. Regarding the PANSS positive and CTQ subscales, the CFA showed excellent fit indexes. The factor loadings of the unidimensional models for each measure are presented in Supplementary Table 1 (Table S1).

3.3. Mediation analysis

The correlation matrix (Table 3) and scatterplots (Supplementary Table 2) indicated a significant linear relationship among the variables of interest. Significant results of the mediation analysis are presented in Fig. 2. Significant paths were found from emotional abuse to ASI total score ($\beta = 0.39$; $p < .001$) and from ASI total score to PANSS positive ($\beta = 0.17$; $p = .019$). The indirect effect of emotional abuse on PANSS positive via the ASI total score was confirmed ($\beta = 0.06$; $p = .041$; CI 95% [0.01, 0.13]). All the path coefficients are displayed in Table 4.

4. Discussion

The primary goal of this study was to assess the existing link between CTA, AS, and psychotic symptoms in individuals affected by SSDs. Additionally, we aimed to examine which CTA dimension might influence more AS mediation to positive symptoms of psychosis, and our findings appear to underscore a noteworthy connection among variables explored. Specifically, we found that a history of childhood emotional abuse is linked to a heightened level of AS which mediates the influence on a more pronounced positive symptomatology during adulthood, namely delusions and hallucinations.

Therefore, our research introduces an innovative viewpoint by identifying a distinct correlation between early traumatic experiences, AS, and psychotic symptoms. These experiences could serve as predictive factors directly associated with the development and persistence of AS and consequently of psychotic manifestations and impairment in individuals with psychosis, even if treated with antipsychotics [46].

If present results innovatively highlight a broader understanding of the interplay between childhood trauma, namely emotional abuse, aberrant salience, and psychotic symptoms, other evidence previously explored associations between single domains. Indeed, De Fazio and colleagues investigated the association between CTA and AS and their impact on clinical variables in schizophrenia identifying a significant AS trait in more than half of the adult patients enrolled [28]. Authors found a straightforward correlation between AS with increased severity of delusions, hallucinatory behavior, mannerisms, and unusual thought content (as measured by the PANSS scale) as well as a history of emotional abuse (according to the CTQ tool), concluding that a personal history of childhood emotional abuse and prior psychiatric hospitalizations are both linked to elevated levels of AS [28]. Still, CTA seems to be specific for different disorders, with different consequences depending on the type of trauma experienced, the environmental and psychopathological variables involved, and the psychiatric diagnosis in adulthood [14]. In detail, in the early stages of life patients with a subsequent diagnosis of bipolar disorder exhibit a higher prevalence of

Table 2
Fit indices for the tested unidimensional models according to the original factor structure.

	χ^2/df	CFI	TLI	RMSEA	SRMR
Threshold for good model fit	≤ 2	≥ 0.95	≥ 0.95	≤ 0.05	≤ 0.05
Threshold for acceptable model fit	≤ 3	≥ 0.90	≥ 0.90	≤ 0.08	≤ 0.08
Second-order-factor model (ASI)	2.0	0.96	0.96	0.07	0.13
One-factor model (PANSS positive)	1.2	0.99	0.99	0.03	0.02
One-factor model (CTQ five subscales)	1.6	0.90	0.90	0.08	0.06

χ^2/df : relative chi-square; CFI: Comparative Fit Index; TLI: Tucker-Lewis Index; RMSEA (90% CI): Root Mean Square Error of Approximation (90% confidence interval); SRMR: Standardized Root Mean Squared Residual.

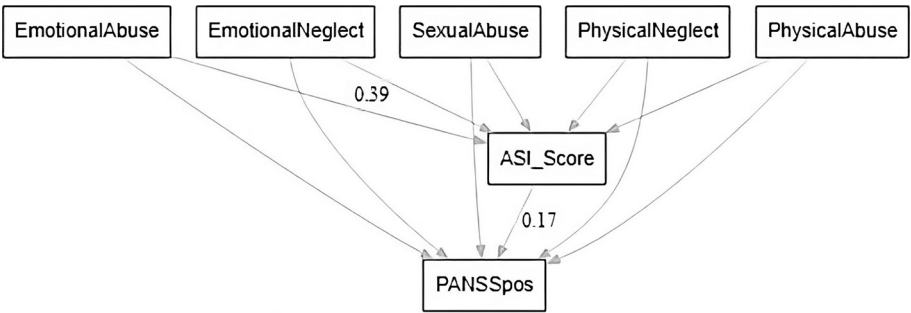


Fig. 2. Graphical representation of the mediation analysis performed between CTQ subscales (independent variables), PANSS positive (dependent variable), and ASI total score (mediator). Only the significant path coefficients are included in the figure.

Table 3
Correlation matrix depicting the relationships among the variables of interest.

	1.	2.	3.	4.	5.	6.	7.
1. PANSS positive	–						
2. Emotional Abuse	0.238**	–					
3. Physical Abuse	0.213**	0.673**	–				
4. Sexual Abuse	0.117	0.315**	0.430**	–			
5. Emotional Neglect	0.135*	0.692**	0.561**	0.144*	–		
6. Physical Neglect	0.140*	0.628**	0.607**	0.139*	0.804**	–	
7. ASI total score	0.233**	0.415**	0.270**	0.149*	0.306**	0.311**	–

PANSS: Positive and negative symptom scale; ASI: Aberrant salience inventory.
** $p < .001$.
* $p < .05$.

Table 4
Parameter estimates of the path diagram model.

Direct effects									
								95% Confidence Interval	
					β	SE	p	Lower	Upper
Emotional Abuse	→	PANSS pos			0.135	0.106	0.164	−0.060	0.357
Physical Abuse	→	PANSS pos			0.117	0.156	0.300	−0.144	0.468
Sexual Abuse	→	PANSS pos			0.008	0.109	0.919	−0.203	0.226
Emotional Neglect	→	PANSS pos			−0.064	0.087	0.551	−0.224	0.119
Physical Neglect	→	PANSS pos			−0.024	0.125	0.819	−0.273	0.216
Indirect effects									
Emotional Abuse	→	ASI Score	→	PANSS pos	0.059	0.032	0.041	0.008	0.127
Physical Abuse	→	ASI Score	→	PANSS pos	−0.012	0.024	0.491	−0.064	0.031
Sexual Abuse	→	ASI Score	→	PANSS pos	0.008	0.020	0.603	−0.028	0.049
Emotional Neglect	→	ASI Score	→	PANSS pos	−0.006	0.016	0.745	−0.037	0.026
Physical Neglect	→	ASI Score	→	PANSS pos	0.022	0.027	0.328	−0.026	0.078
Path coefficients									
ASI Score	→	PANSS pos			0.173	0.047	0.019	0.018	0.213
Emotional Abuse	→	ASI Score			0.391	0.145	<0.001	0.303	0.873
Physical Abuse	→	ASI Score			−0.074	0.206	0.461	−0.555	0.252
Sexual Abuse	→	ASI Score			0.047	0.172	0.591	−0.245	0.431
Emotional Neglect	→	ASI Score			−0.039	0.144	0.742	−0.330	0.235
Physical Neglect	→	ASI Score			0.136	0.210	0.259	−0.174	0.648

PANSS: Positive and negative symptom scale; ASI: Aberrant salience inventory.

maternal absence or loss and economic difficulties, while patients suffering from SSDs display a greater frequency of escape from home, cannabis abuse, psychological abuse, physical abuse, and loneliness, while healthy controls tend to have lower occurrences of paternal absence, neglect of core needs, and significant familial tension [14]. In light of these data, it is not surprising to identify a specific type of CTA among the various risk factors of the psychiatric population, and our data on emotional abuse fits into this context of various forms of adversities that significantly contribute to the development of major psychiatric disorders, with some degree of diagnostic accuracy.

There is literature supporting CTA as a robust predictor of mental illness, tightly intertwined with the gravity of psychiatric symptoms, diminished responsiveness to treatment, prolonged illness duration, exacerbation of patient conditions, and an elevated risk of suicide [47–49]. Specifically, emotional abuse, described as verbal hostility or behavior that jeopardizes the well-being of the child [50], has been already identified as having the most substantial impact in clinical samples [51].

In this regard, the study conducted by Muscatello and colleagues examined potential associations between different categories of CTA and the dimensions of hyperarousal and AS within a group of one hundred adult patients with psychiatric disorders [51]. Although the authors did not specifically enroll a sample of patients affected by SSDs, nor did they correlate with psychotic symptoms clinically significant, they nevertheless confirmed emotional abuse as the only CTA dimension with a role as a predictor of AS.

A possible explanation for these results comes from the increasing recognition of the environment's crucial role in shaping the synaptic

structure of the adult brain, which has directed researchers' attention toward exploring the potential ramifications of educational factors. Additionally, it underscores the significance of traumatic experiences during development, which can lead to enduring neurobiological and neuroendocrine impairments. These enduring effects may contribute to heightened vulnerability in the development of psychopathological disorders [52].

Indeed, while it is well-documented that early trauma can significantly impact a child's development across various domains, including behavior, emotions, social interactions, physical health, and cognitive abilities [53,54], the precise mechanisms through which these harmful effects manifest remain less understood. Some theories suggest that early adversities primarily affect neural pathways related to stress responses and the processing and regulation of emotions, subsequently influencing cognitive control and selective attention [55].

Hence, guided by the idea that abnormal emphasis on significance and heightened salience appear to be plausible factors that mediate or moderate the impact of early trauma as a broad predisposing factor for major psychiatric disorders and their manifestations, our study sought to identify the mediation of AS on CTA experiences in the clinical symptomatology adult patients affected by SSDs.

Another feature that is worth considering when interpreting our results is the aspect relating to cultural factors that could influence the clinical outcomes of psychosis. In fact, if on the one hand, it is easy to hypothesize how cultural background could play a role in the development of AS, including ethnicity, religiousness, education, and personal experiences, strong supporting evidence is still missing [56,57]. However, much more data exists on the role that culture and ethnicity play in

the manifestations of the positive symptoms of schizophrenia, an aspect that may not be captured by our work considering the uniformity of the population involved [58,59]. However, even in this case, there is a lack of data that can evaluate the cultural background in light of confounding factors such as CTA in the onset of psychosis, which therefore should be explored in depth with transcultural and international evaluations.

5. Limitations and strengths

While the innovative findings showcased in this research open up the possibility of better defining the existing intercorrelation among CTA, AS, and psychotic symptomatology, it is important to take into account several limitations when interpreting these results. First, the assessment was conducted through two self-report instruments (i.e., CTQ and ASI) and, although we collected information from participants' personal history about CTA to double-check data consistency, we did not cross-test aberrant salience with convergent clinician-administrated tools (e.g., the Salience Attribution Test – SAT –) [5,60]. In this regard, ASI scores exhibit correlations with psychotic-like experiences and effectively distinguish schizophrenia from other mental health conditions, such as bipolar disorder [34], while the SAT exhibits a sounder construct validity in healthy individuals [61]. Moreover, self-report measures offer the advantage of allowing the feasible enrollment of larger study populations. Indeed, currently, the ASI stands as the sole available assessment tool for gauging the aberrant salience trait [5,19,34], demonstrating notable reliability among individuals with a history of psychosis [19,28,34]. Second, we acknowledge that our sample comprised individuals who were receiving antipsychotic treatments. While this aligns with clinical practice and international treatment guidelines and evidence [62,63], it may potentially influence the expression, as well as the associations, between AS and positive/negative symptoms. This is because antipsychotic medications can inhibit the underlying aberrant dopaminergic activity, and considering the pivotal role of dopamine in salience, it could lead to a reduction in aberrantly salient thoughts and perceptions [6]. Indeed, consistent with Kapur's theoretical framework [6], which posits that AS plays a pivotal role in shaping the subjective experience of psychosis, it is also associated with other assessments of psychotic phenomena, including magical ideation, dissociation, and absorption [64]. Consequently, for a more comprehensive understanding, it is desirable to replicate our findings in a drug-naïve sample, preferably at the onset of the disorder or in an at-high-risk population [3], to mitigate this potential bias. Third, the indirect effect of emotional abuse on PANSS positive via the ASI total score, although significant in statistical terms (i.e., $p = .041$), is close to the standardized limit for significance (i.e., $p = .05$). Therefore, these results should be taken cautiously, and it is hoped that they will be confirmed in the future by different and/or larger studies. Specifically, it would be desirable to replicate this study with a prospective design by enrolling drug-naïve and/or first psychotic episode patients, to limit the confounding effect of antipsychotic therapy or long duration of illness, as well as add a cognitive assessment that can detect the interference of possible cognitive deterioration on AS.

Moreover, we also acknowledge that the cross-sectional design of this study precludes us from drawing firm causal inferences. Given that AS is a dynamic process rooted in dopaminergic biological alterations, personal relationships, and environmental stimuli interactions, it would be beneficial to pursue dynamic follow-up investigations to examine its evolution over time [26]. This result appears to be in line with what was found in our sample, where emotional abuse plays a reinforcing role both on AS and on the positive symptoms, and therefore on the clinical severity, of patients with schizophrenia.

In support of this theory, there is a sufficiently large literature that has identified common childhood risk factors in major psychiatric disorders, and in particular in SSDs [13,14]. For instance, some CTA experiences demonstrate a connection with the clinical severity observed in individuals affected by bipolar disorder and SSD. Notably, early life

adversities linked to abuse and neglect appear to be linked to more pronounced symptom severity compared to those associated with a lack of support or social isolation. This speculation implies that a patient's history of CTA might serve as a valuable tool for identifying those at a heightened risk of experiencing adverse clinical characteristics as we hypnotized as well [13].

6. Conclusion

Our study reveals a noteworthy existing mediation relationship between a specific psychopathological mechanism related to psychosis, such as AS, and a risk factor that transcends diagnostic categories, namely CTA, in their indirect action of maintenance of psychotic symptoms, in particular the positive domain of schizophrenia. Specifically, our findings provide strong support for the primary hypothesis, which suggests that individuals who have experienced emotional abuse during their childhood may exhibit higher levels of AS when they develop psychosis, thus mediating a clinical effect on positive symptoms in adult patients even if treated with antipsychotics.

From a clinical perspective, our study underscores the growing importance of considering early traumatic experiences as potential indicators of disease progression, classification, and staging. This information can be valuable in identifying patients at a heightened risk of experiencing more severe clinical manifestations and an unfavorable course of illness. Conducting a thorough patient history assessment with a focus on identifying CTA and AS traits during psychiatric evaluations of individuals with psychosis, becomes crucial for establishing an appropriate diagnostic and treatment plan.

Conversely, our study also suggests that AS could play a significant role as a mediator between traumatic experiences and the development and persistence of psychotic symptoms. Detecting AS at an early stage in disadvantaged children and adolescents, as well as in adults, may serve as a decisive marker for preventive measures aimed at averting the onset of psychosis.

Further research is warranted to deepen our understanding of the mechanisms through which CTA influences the formation of AS. This includes investigating the origin and the potential existence of both adaptive and aberrant alterations in the salience processing mechanism.

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Ethics approval

The study was conducted in accordance with the principles of the Declaration of Helsinki and approved by the Ethics Committee of University Hospital Mater Domini of Catanzaro (Italy) 'Regione Calabria, sezione Area Centro' (n. 191/2020).

Informed consent statement

Informed consent was obtained from all subjects involved in the study before any procedure took place.

Code availability

The code will be made available upon reasonable request.

CRediT authorship contribution statement

Renato de Filippis: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Matteo Aloï:** Conceptualization, Data curation, Investigation, Methodology, Software, Writing – review & editing. **Marco Tullio Liuzza:** Methodology, Software, Supervision. **Valentina Pugliese:** Data curation, Writing – review & editing. **Elvira Anna Carbone:** Data curation, Writing – review & editing. **Marianna Rania:** Data curation, Writing – review & editing. **Cristina Segura-Garcia:** Data curation, Project administration, Supervision, Validation, Writing – review & editing. **Pasquale De Fazio:** Conceptualization, Data curation, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

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Data availability

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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References

- Novembre G, Pawar VM, Bufacchi RJ, Kilintari M, Srinivasan M, Rothwell JC, et al. Saliency detection as a reactive process: unexpected sensory events evoke Corticomuscular coupling. *J Neurosci* 2018;38:2385–97. <https://doi.org/10.1523/JNEUROSCI.2474-17.2017>.
- Kahnt T, Tobler PN. Reward, Value, and Saliency. *Decis Neurosci*. Elsevier; 2017. p. 109–20. <https://doi.org/10.1016/B978-0-12-805308-9.00009-9>.
- Lisi G, Raballo A, Ribolsi M, Nioiu C, Siracusano A, Preti A. Aberrant salience in adolescents is related to indicators of psychopathology that are relevant in the prodromal phases of psychosis. *Early Interv Psychiatry* 2021;15:856–64. <https://doi.org/10.1111/eip.13022>.
- Palaniyappan L, Mallikarjun P, Joseph V, White TP, Liddle PF. Reality distortion is related to the structure of the salience network in schizophrenia. *Psychol Med* 2011;41:1701–8. <https://doi.org/10.1017/S0033291710002205>.
- Pugliese V, de Filippis R, Aloï M, Rotella P, Carbone EA, Gaetano R, et al. Aberrant salience correlates with psychotic dimensions in outpatients with schizophrenia spectrum disorders. *Ann Gen Psychiatry* 2022;21:25. <https://doi.org/10.1186/s12991-022-00402-5>.
- Kapur S. Psychosis as a state of aberrant salience: a framework linking biology, phenomenology, and pharmacology in schizophrenia. *Am J Psychiatry* 2003;160:13–23. <https://doi.org/10.1176/appi.ajp.160.1.13>.
- Palaniyappan L, Liddle PF. Does the salience network play a cardinal role in psychosis? An emerging hypothesis of insular dysfunction. *J Psychiatry Neurosci* 2012;37:17–27. <https://doi.org/10.1503/jpn.100176>.
- Bromberg-Martin ES, Matsumoto M, Hikosaka O. Dopamine in motivational control: rewarding, aversive, and alerting. *Neuron* 2010;68:815–34. <https://doi.org/10.1016/j.neuron.2010.11.022>.
- Kapur S. How antipsychotics become anti-‘psychotic’ – from dopamine to salience to psychosis. *Trends Pharmacol Sci* 2004;25:402–6. <https://doi.org/10.1016/j.tips.2004.06.005>.
- de Filippis R, Staltari FA, Aloï M, Carbone EA, Rania M, Destefano L, et al. Effectiveness of SGA-LAIs on clinical, cognitive, and social domains in schizophrenia: results from a prospective naturalistic study. *Brain Sci* 2023;13:577. <https://doi.org/10.3390/brainsci13040577>.
- Savva C, Griffin B, Abed R. Towards a unified account of aberrant salience in psychosis: proximate and evolutionary mechanisms. *Evol Psychol Sci* 2024;10:52–69. <https://doi.org/10.1007/s40806-024-00384-5>.
- Rosenfield PJ, Jiang D, Pauselli L. Childhood adversity and psychotic disorders: epidemiological evidence, theoretical models and clinical considerations. *Schizophr Res* 2022;247:55–66. <https://doi.org/10.1016/J.SCHRES.2021.06.005>.
- Carbone EA, Pugliese V, Bruni A, Aloï M, Calabrò G, Jaén-Moreno MJ, et al. Adverse childhood experiences and clinical severity in bipolar disorder and schizophrenia: a transdiagnostic two-step cluster analysis. *J Affect Disord* 2019;259. <https://doi.org/10.1016/j.jad.2019.08.049>.
- Bruni A, Carbone EA, Pugliese V, Aloï M, Calabrò G, Cerminara G, et al. Childhood adversities are different in schizophrenic Spectrum disorders, bipolar disorder and major depressive disorder. *BMC Psychiatry* 2018;18:391. <https://doi.org/10.1186/s12888-018-1972-8>.
- Preti A, Cella M, Raballo A, Vellante M. Psychotic-like or unusual subjective experiences? The role of certainty in the appraisal of the subclinical psychotic phenotype. *Psychiatry Res* 2012;200:669–73. <https://doi.org/10.1016/j.psychres.2012.07.014>.
- Steardo L, Carbone EA, de Filippis R, Pisanu C, Segura-Garcia C, Squassina A, et al. Application of support vector machine on fMRI data as biomarkers in schizophrenia diagnosis: a systematic review. *Front Psych* 2020;11. <https://doi.org/10.3389/fpsy.2020.00588>.
- Jaaro-Peled H, Sawa A. Neurodevelopmental factors in schizophrenia. *Psychiatr Clin North Am* 2020;43:263–74. <https://doi.org/10.1016/j.psc.2020.02.010>.
- Preti A, Cella M, Raballo A. Letter to the editor: how psychotic-like are unusual subjective experiences? *Psychol Med* 2011;41:2235–6. <https://doi.org/10.1017/S0033291711000936>.
- Raballo A, Cicero DC, Kerns JG, Sanna S, Pintus M, Agartz I, et al. Tracking salience in young people: a psychometric field test of the aberrant salience inventory (ASI). *Early Interv Psychiatry* 2019;13:64–72. <https://doi.org/10.1111/eip.12449>.
- Jauhar S, Johnstone M, McKen P. Schizophrenia. *Lancet* 2022;Jan:473–86. [https://doi.org/10.1016/S0140-6736\(21\)01730-X](https://doi.org/10.1016/S0140-6736(21)01730-X).
- de Filippis R, Carbone EA, Gaetano R, Bruni A, Pugliese V, Segura-Garcia C, et al. Machine learning techniques in a structural and functional MRI diagnostic approach in schizophrenia: a systematic review. *Neuropsychiatr Dis Treat* 2019;15:1605–27. <https://doi.org/10.2147/NDT.S202418>.
- La Valle R. Esquizofrenia en la CIE-11: comparación con la CIE-10 y el DSM-5. *Rev Psiquiatr Salud Ment* 2020;13:95–104. <https://doi.org/10.1016/j.rpsm.2020.01.001>.
- Ballerini A, Tortorelli M, Marino P, Appignanesi C, Baschiroto C, Mallardo L, et al. Aberrant salience relationship with first rank symptoms. *Ann Gen Psychiatry* 2022;21:8. <https://doi.org/10.1186/s12991-022-00383-5>.
- Gawęda Ł, Göritz AS, Moritz S. Mediating role of aberrant salience and self-disturbances for the relationship between childhood trauma and psychotic-like experiences in the general population. *Schizophr Res* 2019;206:149–56. <https://doi.org/10.1016/j.schres.2018.11.034>.
- Chun C, Gross G, Mielock A, Kwapił T. Aberrant salience predicts psychotic-like experiences in daily life: An experience sampling study. *Schizophr Res* 2020;220:218–24. <https://doi.org/10.1016/j.schres.2020.03.016>.
- Azzali S, Pelizza L, Scazza I, Paterlini F, Garlassi S, Chiri LR, et al. Examining subjective experience of aberrant salience in young individuals at ultra-high risk (UHR) of psychosis: a 1-year longitudinal study. *Schizophr Res* 2022;241:52–8. <https://doi.org/10.1016/j.schres.2021.12.025>.
- Loewy RL, Corey S, Amirfathi F, Dabit S, Fulford D, Pearson R, et al. Childhood trauma and clinical high risk for psychosis. *Schizophr Res* 2019;205:10–4. <https://doi.org/10.1016/j.schres.2018.05.003>.
- De Fazio P, Pugliese V, Cattolico M, Aloï M, Segura-Garcia C. The relationship between childhood trauma and aberrant salience: a preliminary study in patients with schizophrenia. *J Psychopathol* 2020;26:28–35. <https://doi.org/10.36148/2284-0249-367>.
- American Psychiatric Association. *Diagnostic and statistical manual of mental disorders: DSM-5*. 5th ed. 2013. Washington.
- First M, Williams J, Karg R. *Structured clinical interview for DSM-5 disorders, clinician version (SCID-5-CV)*. Am Psychiatr Assoc 2015.
- World Medical Association. *World medical association declaration of Helsinki: ethical principles for medical research involving human subjects*. *JAMA* 2013;310:2191. <https://doi.org/10.1001/jama.2013.281053>.
- Kline P. *An Easy Guide to Factor Analysis*. 2014. <https://doi.org/10.4324/9781315788135>.
- Gardner DM, Murphy AL, O'Donnell H, Centorrino F, Baldessarini RJ. International consensus study of antipsychotic dosing. *Am J Psychiatry* 2010;167:686–93. <https://doi.org/10.1176/appi.ajp.2009.09060802>.
- Cicero DC, Kerns JG, McCarthy DM. The aberrant salience inventory: a new measure of psychosis proneness. *Psychol Assess* 2010;22:688–701. <https://doi.org/10.1037/a0019913>.
- Raballo A, Scanu R, Petretto DR, Preti A. Aberrant salience and psychosis risk symptoms in Italian undergraduate students: further validation of the aberrant salience inventory. *Early Interv Psychiatry* 2014;8:134.
- Innamorati M, Erbutto D, Venturini P, Fagioli F, Ricci F, Lester D, et al. Factorial validity of the childhood trauma questionnaire in Italian psychiatric patients. *Psychiatry Res* 2016;245:297–302. <https://doi.org/10.1016/j.psychres.2016.08.044>.
- Kay SR, Fiszbein A, Opler LA. The positive and negative syndrome scale (PANSS) for schizophrenia. *Schizophr Bull* 1987;13:261–76. <https://doi.org/10.1093/schbul/13.2.261>.
- Wallwork RS, Fortgang R, Hashimoto R, Weinberger DR, Dickinson D. Searching for a consensus five-factor model of the positive and negative syndrome scale for schizophrenia. *Schizophr Res* 2012;137:246–50. <https://doi.org/10.1016/j.schres.2012.01.031>.
- Widaman KF, Revelle W. Thinking thrice about sum scores, and then some more about measurement and analysis. *Behav Res Methods* 2022;55:788–806. <https://doi.org/10.3758/s13428-022-01849-w>.

- [40] Hu L, Bentler PM. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct Equ Model A Multidiscip J* 1999;6:1–55. <https://doi.org/10.1080/10705519909540118>.
- [41] R Foundation for Statistical Computing. R Core Team. R: A Language and Environment for Statistical Computing. 2021.
- [42] Posit Team. RStudio: Integrated development environment for R. Boston, MA: Posit Software, PBC; 2023.
- [43] Rosseel YL, An R. Package for structural equation modeling. *J Stat Softw* 2012;48:1–36.
- [44] Revelle W. Psych: Procedures for Psychological, Psychometric, and Personality Research. 2023.
- [45] Jorgensen TD, Pornprasertmanit S, Schoemann AM, Rosseel Y. semTools: Useful tools for structural equation modeling. R package version 0.5–6. 2022.
- [46] Pugliese V, de Filippis R, Aloï M, Carbone EA, Rania M, Segura-Garcia C, et al. Cognitive biases are associated with aberrant salience experience in schizophrenia spectrum disorders. *Spanish J Psychiatry Ment Heal* 2023. <https://doi.org/10.1016/j.sjpmh.2023.07.001>.
- [47] Green JG, McLaughlin KA, Berglund PA, Gruber MJ, Sampson NA, Zaslavsky AM, et al. Childhood adversities and adult psychiatric disorders in the National Comorbidity Survey Replication I. *Arch Gen Psychiatry* 2010;67:113. <https://doi.org/10.1001/archgenpsychiatry.2009.186>.
- [48] Hovens JGFM, Giltay EJ, Spinhoven P, van Hemert AM, Penninx BWJH. Impact of childhood life events and childhood trauma on the onset and recurrence of depressive and anxiety disorders. *J Clin Psychiatry* 2015;76:931–8. <https://doi.org/10.4088/JCP.14m09135>.
- [49] McLaughlin KA, Green JG, Gruber MJ, Sampson NA, Zaslavsky AM, Kessler RC. Childhood adversities and adult psychiatric disorders in the National Comorbidity Survey Replication II. *Arch Gen Psychiatry* 2010;67:124. <https://doi.org/10.1001/archgenpsychiatry.2009.187>.
- [50] Bremner JD, Bolus R, Mayer EA. Psychometric properties of the early trauma Inventory–Self Report. *J Nerv Ment Dis* 2007;195:211–8. <https://doi.org/10.1097/01.nmd.0000243824.84651.6c>.
- [51] Muscatello MRA, Rizzo A, Celebre L, Mento C, Pandolfo G, Cedro C, et al. The wounds of childhood: early trauma subtypes, salience and hyperarousal in a sample of adult psychiatric patients. *Int J Soc Psychiatry* 2020;66:3–9. <https://doi.org/10.1177/0020764019872227>.
- [52] Von Werne Baes C, de Carvalho Tofoli SM, Martins CMS, Jurueña MF. Assessment of the hypothalamic–pituitary–adrenal axis activity: glucocorticoid receptor and mineralocorticoid receptor function in depression with early life stress – a systematic review. *Acta Neuropsychiatr* 2012;24:4–15. <https://doi.org/10.1111/j.1601-5215.2011.00610.x>.
- [53] Bremner JD, Vermetten E. Stress and development: behavioral and biological consequences. *Dev Psychopathol* 2001;13:473–89. <https://doi.org/10.1017/S0954579401003042>.
- [54] De Filippis R, Carbone EA, D'Angelo M, Liuzza MT, De Fazio P, Steardo L. The mediation role of impulsivity between childhood trauma and dissociative symptomatology in bipolar disorder. *Riv Psichiatr* 2023;58:84–92. <https://doi.org/10.1708/4022.39978>.
- [55] Lim L, Hart H, Mehta MA, Simmons A, Mirza K, Rubia K. Neural correlates of error processing in young people with a history of severe childhood abuse: An fMRI study. *Am J Psychiatry* 2015;172:892–900. <https://doi.org/10.1176/appi.ajp.2015.14081042>.
- [56] Shoib S, Das S, Gupta AK, Ullah I, Javed S, Nocera A, et al. Religion, spirituality, and coping among the psychiatric population: a narrative review. *Open J Psychiatry Allied Sci* 2024.
- [57] Cicero DC, Cohn JR. The role of ethnic identity, self-concept, and aberrant salience in psychotic-like experiences. *Cultur Divers Ethnic Minor Psychol* 2018;24:101–11. <https://doi.org/10.1037/cdp0000160>.
- [58] Viswanath B, Chaturvedi SK. Cultural aspects of major mental disorders: a critical review from an Indian perspective. *Indian J Psychol Med* 2012;34:306–12. <https://doi.org/10.4103/0253-7176.108193>.
- [59] Ghanem M, Evangeli-Dawson C, Georgiades A. The role of culture on the phenomenology of hallucinations and delusions, explanatory models, and help-seeking attitudes: a narrative review. *Early Interv Psychiatry* 2023;17:843–63. <https://doi.org/10.1111/eip.13449>.
- [60] Roiser JP, Stephan KE, Den Ouden HEM, Barnes TRE, Friston KJ, Joyce EM. Do patients with schizophrenia exhibit aberrant salience? *Psychol Med* 2009;39:199–209. <https://doi.org/10.1017/S0033291708003863>.
- [61] Schmidt K, Roiser JP. Assessing the construct validity of aberrant salience. *Front Behav Neurosci* 2009;3. <https://doi.org/10.3389/NEURO.08.058.2009>.
- [62] de Filippis R, De Fazio P, Gaetano R, Steardo L, Cedro C, Bruno A, et al. Current and emerging long-acting antipsychotics for the treatment of schizophrenia. *Expert Opin Drug Saf* 2021;20:771–90. <https://doi.org/10.1080/14740338.2021.1910674>.
- [63] Magliocco F, de Filippis R, Aloï M, Staltari FA, Gaetano R, Segura-Garcia C, et al. Second-generation long-acting injections anti-psychotics improve executive functions in patients with schizophrenia: a 12-month real-world study. *Int J Psychiatry Clin Pract* 2020;24:201–7. <https://doi.org/10.1080/13651501.2020.1737134>.
- [64] Van Der Gaag M. A neuropsychiatric model of biological and psychological processes in the remission of delusions and auditory hallucinations. *Schizophr Bull* 2006;32(Suppl):1. <https://doi.org/10.1093/SCHBUL/SBL027>.